Experiment No 11

Create a MongoDB Atlas cluster, connect it to a web application, and perform basic CRUD operations

Required Softwares/tools:

Software	**Purpose**
Node.js	Runs your server.js
VS Code	Write/edit server.js and index.html
MongoDB Atlas (online account)	Stores your student records online. It will act as a database.
Web Browser	Opens your application and runs it.
npm (Node Package Manager)	Installs Node.js packages: express, mongodb and dotenv

```
E:\Btech SIT\Web App Development>cd "HTML Practical"

E:\Btech SIT\Web App Development\HTML Practical>cd Exp11

E:\Btech SIT\Web App Development\HTML Practical\Exp11>npm init -y
Wrote to E:\Btech SIT\Web App Development\HTML Practical\Exp11\package.json:

{
  "name": "exp11",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

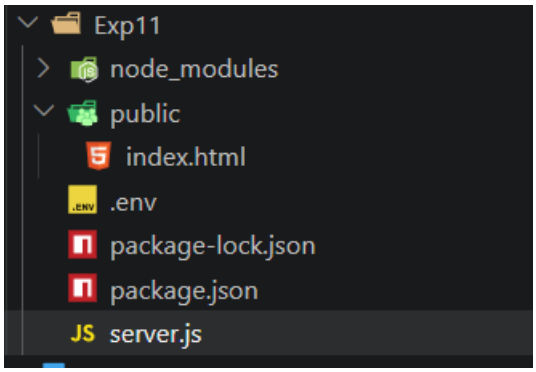
```
E:\Btech SIT\Web App Development\HTML Practical\Exp11>npm install express mongodb dotenv

added 81 packages, and audited 82 packages in 12s

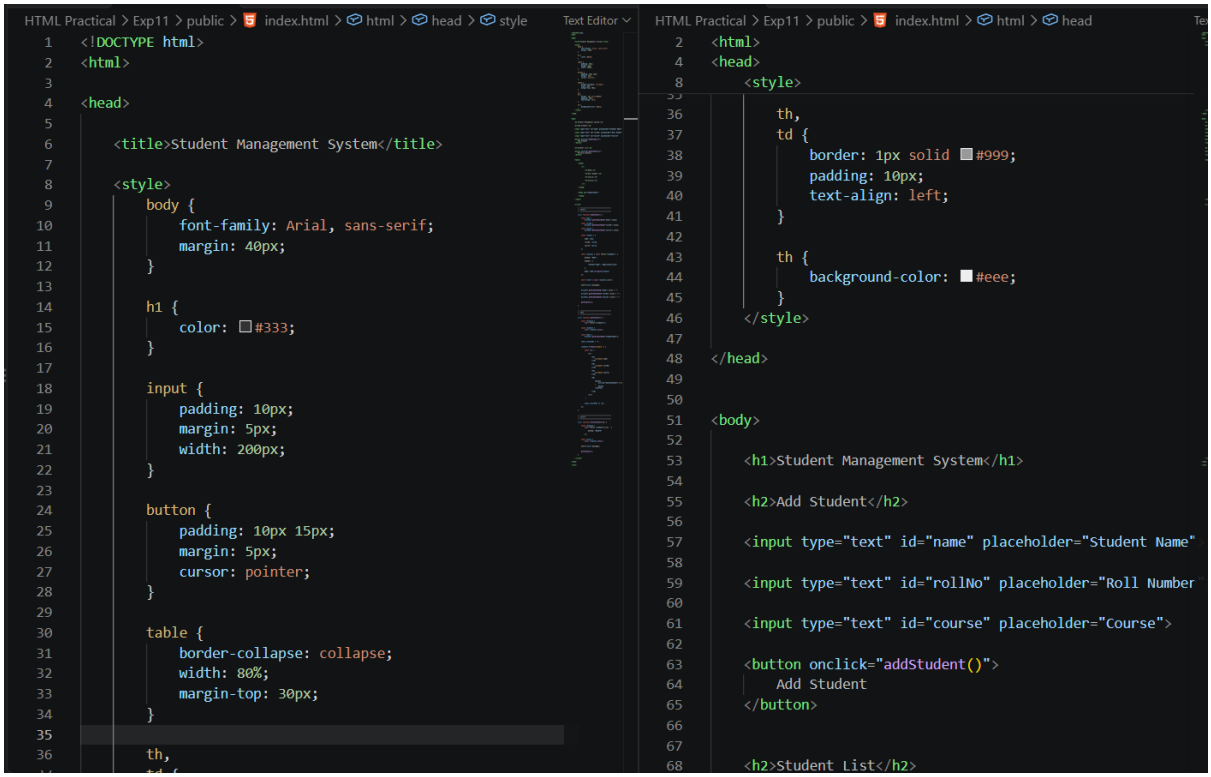
29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Folder Structure :-



Index.html code :-



```
<!DOCTYPE html>
<html>

<head>

    <title>Student Management System</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 40px;
        }

        h1 {
            color: #333;
        }

        input {
            padding: 10px;
            margin: 5px;
            width: 200px;
        }

        button {
            padding: 10px 15px;
            margin: 5px;
            cursor: pointer;
        }

        table {
            border-collapse: collapse;
            width: 80%;
            margin-top: 30px;
        }

        th,
        td {
            border: 1px solid #999;
            padding: 10px;
            text-align: left;
        }

        th {
            background-color: #eee;
        }
    </style>

</head>


<body>

    <h1>Student Management System</h1>

    <h2>Add Student</h2>

    <input type="text" id="name" placeholder="Student Name"

    <input type="text" id="rollNo" placeholder="Roll Number

    <input type="text" id="course" placeholder="Course">

    <button onclick="addStudent()">
        Add Student
    </button>


    <h2>Student List</h2>
```

Name :- Parth Bhayani Prn No :- 26070122512 Batch :- B3



```
<html>
<body>
    <button onclick="getStudents()">
        Refresh Students
    </button>

    <table>

        <thead>

            <tr>

                <th>Name</th>

                <th>Roll Number</th>

                <th>Course</th>

                <th>Action</th>

            </tr>

        </thead>

        <tbody id="studentTable">

        </tbody>

    </table>

    <script>

        // ==================================
```



```
<html>
<body>
    <script>

        // CREATE
        // ==================================

        async function addStudent() {

            const name =
                document.getElementById("name").value;

            const rollNo =
                document.getElementById("rollNo").value;

            const course =
                document.getElementById("course").value;


            const student = {

                name: name,

                rollNo: rollNo,

                course: course

            };


            const response = await fetch("/students", {

                method: "POST",

                headers: {

                    "Content-Type": "application/json"
```



```
<html>
<body>
    <script>
        async function addStudent() {
            const response = await fetch("/students", {
                },

                body: JSON.stringify(student)

            });


            const result = await response.json();


            alert(result.message);


            document.getElementById("name").value = "";

            document.getElementById("rollNo").value = "";

            document.getElementById("course").value = "";


            getStudents();

        }


        // ==================================
        // READ
        // ==================================

        async function getStudents() {
```



```
<html>
<body>
    <script>
        async function getStudents() {
            const response =
                await fetch("/students");


            const students =
                await response.json();


            const table =
                document.getElementById("studentTable");


            table.innerHTML = "";


            students.forEach(student => {

                const row = `

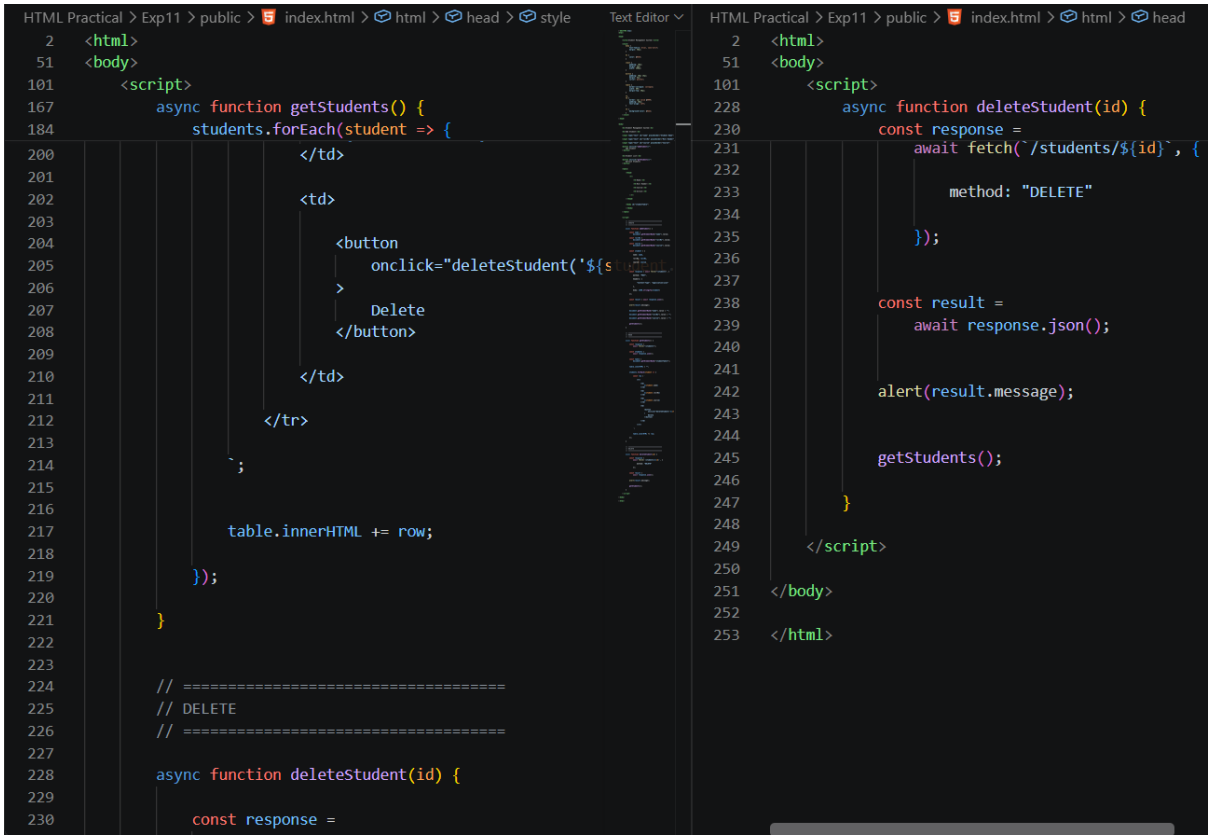
                    <tr>

                        <td>
                            ${student.name}
                        </td>

                        <td>
                            ${student.rollNo}
                        </td>

                        <td>
                            ${student.course}
                        </td>
```

Name :- Parth Bhayani Prn No :- 26070122512 Batch :- B3



```
HTML Practical > Exp11 > public > index.html > html > head > style
  <html>
  <body>
      <script>
          async function getStudents() {
              students.forEach(student => {
                          </td>

                          <td>

                              <button
                                  onclick="deleteStudent('${s
                              >
                                  Delete
                              </button>

                          </td>

                      </tr>

                  `;


                  table.innerHTML += row;

              });

          }


          // ====================================
          // DELETE
          // ====================================

          async function deleteStudent(id) {

              const response =
```

```
HTML Practical > Exp11 > public > index.html > html > head
  <html>
  <body>
      <script>
          async function deleteStudent(id) {
              const response =
                  await fetch(`/students/${id}`, {

                      method: "DELETE"

                  });


              const result =
                  await response.json();


              alert(result.message);


              getStudents();

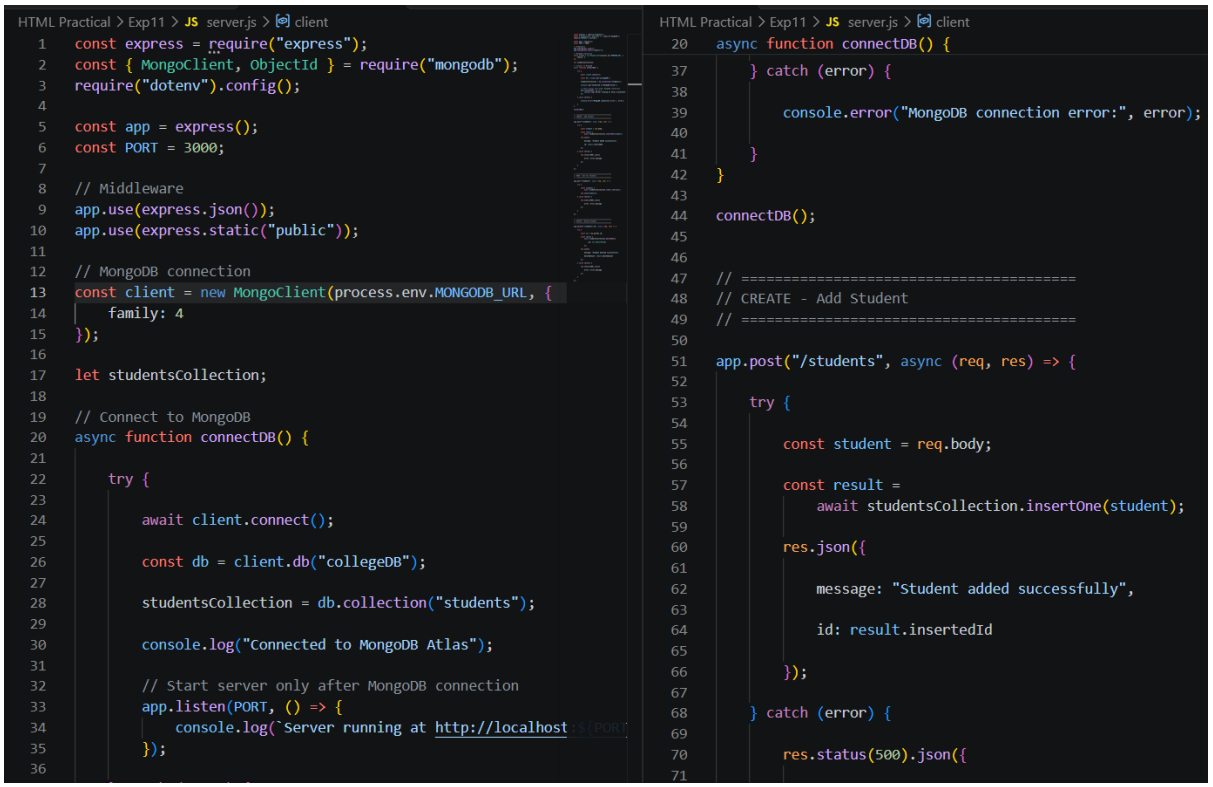
          }

      </script>

  </body>

  </html>
```

Script.js Code :-



```
HTML Practical > Exp11 > JS server.js > client
  const express = require("express");
  const { MongoClient, ObjectId } = require("mongodb");
  require("dotenv").config();

  const app = express();
  const PORT = 3000;

  // Middleware
  app.use(express.json());
  app.use(express.static("public"));

  // MongoDB connection
  const client = new MongoClient(process.env.MONGODB_URL, {
      family: 4
  });

  let studentsCollection;

  // Connect to MongoDB
  async function connectDB() {

      try {

          await client.connect();

          const db = client.db("collegeDB");

          studentsCollection = db.collection("students");

          console.log("Connected to MongoDB Atlas");

          // Start server only after MongoDB connection
          app.listen(PORT, () => {
              console.log(`Server running at http://localhost
          });
```

```
HTML Practical > Exp11 > JS server.js > client
  async function connectDB() {
      } catch (error) {

          console.error("MongoDB connection error:", error);

      }
  }

  connectDB();


  // ========================================
  // CREATE - Add Student
  // ========================================

  app.post("/students", async (req, res) => {

      try {

          const student = req.body;

          const result =
              await studentsCollection.insertOne(student);

          res.json({

              message: "Student added successfully",

              id: result.insertedId

          });

      } catch (error) {

          res.status(500).json({
```

Name :- Parth Bhayani Prn No :- 26070122512 Batch :- B3

HTML Practical > Exp11 > JS server.js > client

```
51  app.post("/students", async (req, res) => {
72              error: error.message
73
74          });
75
76      }
77
78  });
79
80
81  // ======================================
82  // READ - Get All Students
83  // ======================================
84
85  app.get("/students", async (req, res) => {
86
87      try {
88
89          const students =
90              await studentsCollection.find().toArray();
91
92          res.json(students);
93
94      } catch (error) {
95
96          res.status(500).json({
97
98              error: error.message
99
100         });
101
102     }
103
104 });
105
106
```

HTML Practical > Exp11 > JS server.js > client

```
107 // ======================================
108 // DELETE - Delete Student
109 // ======================================
110
111 app.delete("/students/:id", async (req, res) => {
112
113     try {
114
115         const id = req.params.id;
116
117         const result =
118             await studentsCollection.deleteOne({
119
120                 _id: new ObjectId(id)
121
122             });
123
124         res.json({
125
126             message: "Student deleted successfully",
127
128             deletedCount: result.deletedCount
129
130         });
131
132     } catch (error) {
133
134         res.status(500).json({
135
136             error: error.message
137
138         });
139
140     }
141
142 });
```

Output :-

```
E:\Btech SIT\Web App Development\HTML Practical\Exp11>node server.js
◇ injected env (1) from .env
Connected to MongoDB Atlas
Server running at http://localhost:3000
```

